SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

May 2024

Course Code: **MGT213**

Course: **SERVICES SCIENCE & SERVICE OPERATIONAL MANAGEMENT**

QP No. **:U426-8**

Duration: **3** hours

Max. Marks:**100**

PART - A

Answer all the questions **10 x 2 = 20 Marks**

1. Recall what is service operations.
2. Infer service encounter triad.
3. Explain the transformation of GDL to SDL.
4. Differentiate Customer journey and service blueprint.
5. Discuss any two types of service guarantee with examples.
6. Compare Vehicle Routing and Scheduling.
7. What do you mean by Customer Value Equation?
8. Classify the various service quality gaps.
9. Define Service Decoupling.
10. What is walk through audit?

PART – B

Answer any FOUR questions **4 x 15 = 60 Marks**

11. a. Explain the components of Service Package of Airlines Industry.(8)
    b. Explain the role of technology in services. (7)

12. Discuss the common reasons behind the failure of new services, and what factors contribute to the successful development of innovative offerings with appropriate examples?

13. Examine the five dimensions of service quality and discuss their implications for service firms such as hotels/ hospitals/ banks. How do these dimensions manifest differently in the context of the specific industry, considering examples from hospitality, healthcare, and financial services?

14. Discuss various types of forecasting methods that can be employed to predict customer demand accurately? Consider practical applications of methods like time series analysis, regression analysis, and qualitative techniques to forecast factors such as daily foot traffic, peak dining hours, and menu popularity etc.

15. Consider a global coffee chain like Starbucks, which is renowned for its focus on customer experience and service innovation. Applying the stages of the service design thinking process to Starbucks, how can the company effectively navigate through each stage to enhance its service offerings?

16. Given the increasing instances of abusive behavior towards drivers and opportunistic practices such as fare manipulation or fraudulent complaints, how can Uber implement effective strategies to deter such conduct while maintaining a positive and safe environment for both drivers and passengers?

PART – C

Answer the following **1 x 20 = 20 Marks**

17. a. As the marketing manager of a leading e-commerce platform like Flipkart in India, how can you leverage the gap model for service quality? Provide strategies to address each gap effectively.
    b. You've been appointed as the marketing manager of a renowned hospitality chain like Taj Hotels in India. How would you utilize the SERVQUAL method to ensure the highest level of customer satisfaction? Outline key factors to consider in meeting and exceeding customer expectations.

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End Semester Examinations

May 2025

Course Code: **MGT213**

Course: **SERVICES SCIENCE & SERVICE OPERATIONAL MANAGEMENT**

QP No. **:U153-8**

Duration: 3 hours

Max. Marks:**100**

PART – A

Answer any Four questions **4 x 20 = 80 Marks**

1. Discuss the nature of service and service encounters.
2. Elucidate New Service Development.
3. Elaborate Complaint handling and Service recovery.
4. Explain the Balancing demand and capacity.
5. Evaluate the Vehicle Routing Problem.
6. Develop the strategies for Building a world class service organization.

PART – B

Answer the following **1 x 20 = 20 Marks**

7. Examine the GDP growth in Indian Service sector.

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End Semester Examinations

May 2024

Course Code: **INT422**

Course: **IT PROJECT MANAGEMENT**

QP No. **:U299-8**

Duration: **3** hours

Max. Marks**:100**

PART – A

Answer any FOUR questions **4 x 20 = 80 Marks**

1. a) Define software project management, with its elements. (6)

   b) Justify the need for conducting feasibility studies and explain the steps in conducting feasibility studies. (14)

2. How to identify the software project? On successful identification, explain the process of software project formulation.

3. Distinguish the difference between PERT and CPM. Explain the Network diagram components with the rules for drawing Network Diagram with an example.

4. You are asked to develop an E-Commerce website "cricket.com" which works like Amazon and Flipkart where you can sell all the accessories related to cricket. You can sell Bat, ball, stumps, cap, jersey, shoes, and gloves, etc.. How to do the cost control for this project and how to schedule the project using PERT and CPM.

5. Define Agile Project Management and key principles in Agile. Explain various Agile Methodologies used in Project Management.

6. Explain the key principles of DevOps. How to manage the source code and managing builds using DevOps?

PART – B

Answer the following **1 x 20 = 20 Marks**

7. a) How to perform Software Project audit? When we need to decide Project Termination? Explain the various factors involved in audit and termination. (15)

   b) Write short notes on any two other agile models with their purpose. (5)

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End Semester Examinations

May 2025

Course Code: **INT422**

Course: **IT PROJECT MANAGEMENT**

QP No. **:U056-8** Duration: **3** hours

Max. Marks**:100**

PART – A

Answer any Four questions **4 x 20 = 80 Marks**

1. a) Analyze the key components and methods used in software project cost estimation.

   b) Evaluate various financial appraisal techniques for software projects, discussing their strengths, limitations, and overall impact on project decision-making.

2. Construct a network diagram of your choice, incorporating each of the 12 terminologies of the Critical Path Method. Analyze how these terminologies function within your diagram and list the fundamental rules for drawing a network diagram, explaining their importance in effective project planning and management.

3. Describe the following project management activities to demonstrate your understanding, providing short notes on each:
   a) Risk Analysis and its elements. List all elements and explain any 5.

   b) Project Termination with its reasons, termination approaches, termination activities along with Post-implementation Review.

4. List the various roles in Scrum and explain their responsibilities in detail with diagram. Explain about various terminologies used in scrum with definition.

5. How to control various versions of source codes using DevOps? List down their benefits and various tools for version control. Explain how build automations in DevOps functioning and list its benefits.

6. Write short notes on the following other Agile models:
   a) Extreme Programming.
   b) Feature Driven Development.
   c) Dynamic System Development Model.
   d) Crystal.

PART – B

Answer the following **1 x 20 = 20 Marks**

7. A software company is tasked to complete a Mobile Application Project consisting of the following six tasks with respective dependencies, durations, and costs:

Task	Activity	Predecessors	Duration (Days)	Normal Cost ($)	Crash Duration (Days)	Crash Cost ($)
A	Planning	–	4	1000	3	1400
B	Design	A	5	2000	3	2600
C	Development	B	7	4000	5	5500
D	Testing	C	3	1500	2	2100
E	Documentation	B	4	1200	3	1600
F	Deployment	D, E	2	1000	1	1300

Tasks:

a) Construct the Activity-on-Node (AON) network diagram.

b) Identify the critical path and its duration using CPM.

c) Calculate the Expected Time (T_E) for critical activities using PERT:
   - Optimistic Time (O) = 0.75 × Baseline Duration.
   - Most Likely Time (M) = Baseline Duration.
   - Pessimistic Time (P) = 1.5 × Baseline Duration.

d) Crash all critical path activities by 1 day and compute the new project duration and total cost.

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End Semester Examinations

May 2024

Course Code: **MGT214**

Course: **MARKETING RESEARCH & MARKETING MANAGEMENT**

QP No. **:U318-8**

Duration: **3** hours

Max. Marks**:100**

PART - A

Answer all the questions **10 x 2 = 20 Marks**

1. Define Marketing Mix.
2. Differentiate Consumer Marketing and Services marketing.
3. Discuss the significance of retailing in the contemporary era.
4. Relate the impact of advertising on the consumer's decision-making process.
5. List out the different stages in PLC.
6. List out the 5Ms of Advertising.
7. State the Marketing Research process.
8. Explain the different methods of data collection.
9. List the factors that influence the Consumer's decision-making process.

10. Explain the STP process in marketing. Provide examples of successful marketing campaigns based on effective market segmentation.

PART - B

Answer any Four questions **4 x 15 = 60 Marks**

11. Elucidate the significance of all the external factors that affect the marketing strategies of the firm.

12. Examine factor and multivariate analysis. What are the factors to be considered while drafting a marketing research report?

13. Analyze the different Marketing Research methods with relevant examples. Compare and contrast quantitative and qualitative research methods in marketing. Provide examples of situations where each method is most effective.

14. Evaluate the role of pricing strategies in marketing. How can businesses determine optimal pricing models that balance profitability with customer value perception?

15. Examine the impact of social media on contemporary marketing. How can businesses leverage social platforms to enhance their brand presence and engage with customers?

16. Explore the relationship between marketing and consumer behavior. How do psychological factors influence purchasing decisions, and how can businesses use this understanding to their advantage?

PART - C

Answer the following **1 x 20 = 20 Marks**

17. a) Examine the future trends in marketing and their implications for businesses. How might advancements in technology, consumer

behavior, and global dynamics shape the future of marketing strategies?

b) Evaluate the significance of customer relationship management (CRM) in modern marketing. How can businesses build and maintain strong relationships with their customers?

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SASTRA DEEMED UNIVERSITY

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End Semester Examinations **May 2025**

Course Code: **MGT214**

Course: **MARKETING RESEARCH & MARKETING MANAGEMENT**

QP No. **:U216-8** Duration: **3** hours

Max. Marks**:100**

PART - A

Answer any Four questions **4 x 20 = 80 Marks**

1. Discuss the importance of the "marketing mix" and enumerate its influences towards consumers in competitive markets? Can digital transformation reshape traditional marketing mix strategies? Provide examples of companies effectively adapting their marketing mix to changing consumer preferences.

2. Explain the key stages of "New Product Development (NPD)" . How do companies ensure market success at each stage through various strategies? Explain with examples about the different products that have successfully navigated the NPD process.

3. Discuss the features of different elements of the "promotion mix" and justify how promotion mix elements, and their integrated approach is essential for the brand success? Explain with examples about different product or companies that have successfully leveraged the promotion mix.

4. Elucidate the key steps in the "marketing research process", and how do they help businesses to make informed and scientific decisions? Describe how different types of marketing research are crucial for understanding consumer needs and market trends with examples?

5. Describe the different "sampling methods" used in research, and how do they impact the accuracy of findings? Choosing the right sampling technique crucial for reliable and valid results -Justify

6. Discuss the key components of "internet marketing" and elucidate how internet marketing has transformed the traditional marketing strategies, for customer engagement with examples?

PART - B

Answer the following **1 x 20 = 20 Marks**

7. a) Select one FMCG product that interests you and design a market survey to analyze the key factors influencing consumer behavior.

   b) Develop an advertising plan using the 4Ps (Product, Price, Place, Promotion) to strategically position the product in the market and enhance consumer engagement. How can these strategies improve brand perception and drive consumer loyalty?

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SASTRA DEEMED UNIVERSITY

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End Semester Examinations

May 2025

Course Code: **INT434**

Course: **ADVANCED SOCIAL, TEXT AND MEDIA ANALYTICS**

QP No.: **U103-8**

Duration: **3** hours

Max. Marks: **100**

PART - A

Answer any FOUR questions **4 x 20 = 80 Marks**

1. a) Using the Interquartile Range (IQR) method, describe the steps involved in detecting outliers in a given dataset. (12)
   Illustrate your explanation with the following dataset:
   3, 7, 9, 12, 15, 18, 21, 25, 30, 35.
   i) Calculate of Q1, Q3, and IQR.
   ii) Find the lower and upper bounds for outlier detection.
   iii) Identify of any outliers in the dataset.

   b) Summarize the core text mining operations. (8)

2. Given a social media platform where users post either textual posts (T) or image/video posts (I), their sentiment can be classified as Positive (P) or Negative (N). The initial probabilities are P(P) = 0.6 and P(N) = 0.4, with transition probabilities P(P → P) = 0.7, P(P → N) = 0.3, P(N → N) = 0.6, and P(N → P) = 0.4. The emission probabilities are P(T|P) = 0.8, P(I|P) = 0.2, P(T|N) = 0.4, and P(I|N) = 0.6. Given the observation sequence [T, I, I],

a) Calculate the probability of the observation sequence using the Forward Algorithm. (10)

b) Using the Viterbi Algorithm, find the most likely sequence of weather states. (10)

3. a) Perform the G-test for the following data to determine if the observed frequencies differ significantly from the expected frequencies: Observed frequencies (O): [8, 10, 12, 15, 7, 8], Expected frequencies (E): [10, 10, 10, 10, 10, 10], Significance level (α): 0.05, Chi-square critical value for degrees of freedom (df=5 at α=0.05 is11.070). Compare the computed G statistic with the critical value 11.070 and provide the conclusion based on the comparison. (10)

b) Given the following drop-off counts: Homepage: 250 Product Page: 300 Cart Page: 200 Checkout Page: 50 and Total Sessions: 1000. (5+5)

i) Calculate the drop-off rate and cumulative drop-off rate for each page as a percentage of the total sessions.

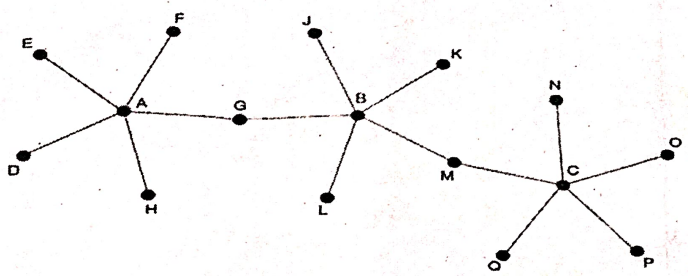
ii) Identify the page contributing the most to drop-offs.

4. Write short notes on: (5+5+5+5)
   a) A/B testing
   b) On-page SEO Vs. Off-page SEO
   c) Web crawling and Indexing
   d) Web traffic model.

5. Given the following graph: Vertices: A, B, C, D, E, F, Edges: AB, BC, BE, BD, CD, CE, DE, DF.

a) Compute the k-core collapse sequence for k=1,2,3, and 4. (10)

b) Calculate local and global centrality for the following graph.(10)



6. For the given directed graph with 4 nodes (A, B, C, D) and the following edges: A → B, A → C, B → C, C → D, D → A. Apply the PageRank algorithm to calculate the PageRank values for the nodes A, B, C, and D. [Hint: do only 5 iterations]

PART - B

Answer the following **1 x 20 = 20 Marks**

7. a) Discuss the importance of link analysis and random graph models in social network analysis. (10)

b) Calculate entropy and purity for the given group of clusters. (10)

Cluster Id:	A	B	C
TOPIC 1:	250	20	10
TOPIC 2:	20	180	80
TOPIC 3:	30	100	210

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P	~P
10	90

For HAS

	SP	S~P	~SP	~S~P
H	92	85	90	5
~H	8	15	10	95

For A

	H	~H
A	90	5
~A	10	95

For B

	H	~H
B	92	90
~B	8	10

b) Explain the difference in the terms
i) Web crawling and indexing.
ii) Affiliation and identity in social context.

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End Semester Examinations

Dec 2023

Course Code: **INT434**

Course: **ADVANCED SOCIAL, TEXT AND MEDIA ANALYTICS**

QP No. **:UD005-7**

Duration: **3** hours

Max. Marks**:100**

PART – A

Answer any Four questions **4 x 20 = 80 Marks**

1. Explain with a detailed note on purpose of classification, and also elaborate KNN classification algorithm, its efficiency, draw backs and perform the same on the following data set and predict the class for the following.
   a) Subject1=7, Subject2=9.
   b) Subject1= 7, Subject2 =7.
   c) Subject1 = 6, Subject2=3.
   d) Subject1=4, Subject2=5.

Subject1	Subject2	RESULT
4	3	Fail
6	7	Pass
7	6	Pass
5	4	Fail
9	9	Pass

2. a) Making use of the given data set apply Naive Bayes algorithm and predict that if a person has the following attributes then which type of race he is
   Race = {Attribute1, Attribute2, Attribute3} (10)

Race	Attribute1	Attribute2	Attribute3	total
South Asian	35	45	0	65
Mongolians	40	30	35	40
Europeans	5	10	5	15
total	80	85	40	120

b) Illustrate the role and importance of Apriori algorithm and apply the same to measure the associations between data items. (10)
Min. Support=45%
Threshold confidence = 70%

Transactions	Items
1	A1, A3, A4
2	A2, A3, A5
3	A1, A2,A3,A5
4	A3, A4, A5
5	A2, A5

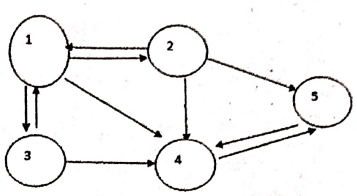
3. a) Explain NLP in detail the 1) Components 2) levels/phases 3) IR with a suitable real time example. (10)

b) Write with suitable example the concept and working principle behind topic detection. (10)

4. a) Apply Agglomerative Hierarchical clustering after explaining the same for the following set of one-dimensional data points 17,20,21,25,33,27,43,44,40,44,38

b) Write in detail any one probabilistic model for information extraction

5. Making use of Matrix method and also compare the result with numerical formula method for the given graph below find the page rank of all the nodes after running the page rank algorithm for three iterations with teleportation factor Beta(B)=0.8

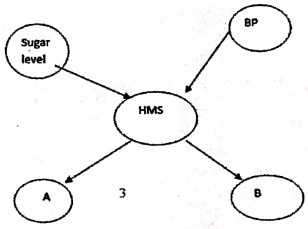


6. Explain the four types of social media analytics with suitable examples.

PART – B

Answer the following **1 x 20 = 20 Marks**

7. a) In the following system depicted find
   i) P (A, ~H, ~S, ~P)
   ii) P(A,B,H,S,~P)



For sugar level

S	~S
2	98

For BP